Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 17)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

Table 1

Site	pH	Dissolved zinc / mg dm^{-3}	Dissolved oxygen / % saturation	Number of invertebrate taxa	Mayfly larvae present?
Upstream	7.1	0.02	96	18	Yes
50 m downstream	3.4	6.5	94	3	No
2 km downstream	5.9	1.1	95	9	No

0 1 . 1 Explain how the acidic drainage water is produced in the spoil heap.

[3 marks]

0 1 . 2 Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

[3 marks]

0 1 . 3 Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

[2 marks]

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0 1 . 4 Describe how acid mine drainage can be controlled.

[2 marks]

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0 2 Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 2 . 1 Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 2 . 2 Describe **two** other features of aircraft design that reduce noise.

[2 marks]

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2 .....

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0 2 . 3 Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle .....

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Night flight restrictions .....

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Noise-based landing charges / quota count .....

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Multiple runways .....

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0 2 . 4 Explain why aircraft are noisier on take-off than on landing.

[1 mark]

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0 3 . 1 Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

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0 3 . 2 Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

0 3 . 3 Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

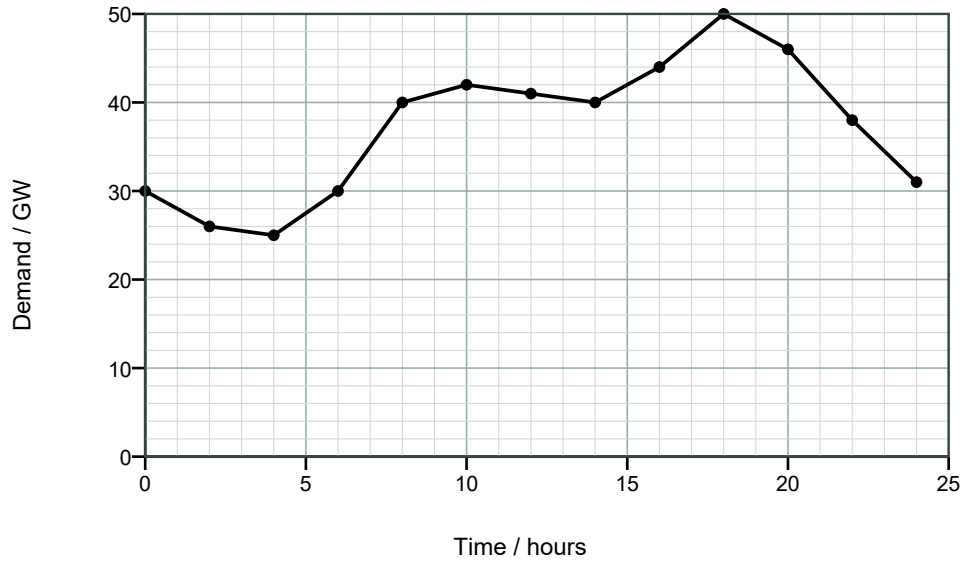
0 3 . 4 Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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0 4 Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



0 4 . 1 Describe the pattern of demand shown in Figure 1.

[2 marks]

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0 4 . 2 Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

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0 4 . 3 Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

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0 4 . 4 Give **two** causes of fluctuations in energy **supply**.

[2 marks]

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0 5 . 1 Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

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0 5 . 2 Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

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0 5 . 3 Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

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0 5 . 4 Describe how deep ocean currents are monitored.

[1 mark]

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0 6 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 6 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 6 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 7 . 1 Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

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0 7 . 2 State the sources of the two fuels used in fusion research.

[2 marks]

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0 7 . 3 Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

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0 7 . 4 Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

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0 8 . 1 Explain **three** advantages of biofuels over other renewable energy resources.

[3 marks]

0 8 . 2 Explain **three** disadvantages of biofuel crops.

[3 marks]

0 8 . 3 Describe **two** new biofuel technologies.

[2 marks]

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0 8 . 4 Explain how new low-temperature turbine technology could allow geothermal electricity generation in the UK.

[2 marks]

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0 9 . 1 Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 9 . 2 Describe **four** principles of designing vehicles for end of life.

[4 marks]

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0 9 . 3 Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

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0 9 . 4 Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

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1 0 . 1 As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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1 0 . 2 State what is meant by the **stock** (resource base) of a mineral.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS